Video Transcript - Part 13

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Hey everyone, welcome to part thirteen. In this video we're covering three services that work closely together: AWS Glue, Glue Crawlers, and Amazon Athena. Together they solve a very common problem. You have lots of raw data sitting in S3, and you want to understand it, prepare it, and ask questions about it, without managing any servers.

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Here's the plan. We'll start with the big picture of how these services fit together. Then we'll look at what AWS Glue is, the Data Catalog, crawlers, and ETL jobs. After that we'll move on to Amazon Athena, and some tips for keeping your queries fast and cheap. We'll finish with a quick guide on when to use which.

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Let's start with the big picture. Imagine a bucket in S3 full of files, CSVs, JSON, maybe Parquet. The files are just data. Nothing tells you what columns they have or what type each column is. A Glue crawler scans that data and works out the structure. It saves what it finds in the Glue Data Catalog as table definitions. Then Amazon Athena uses those tables, so you can query the files with plain SQL. Glue discovers and organizes, and Athena queries. This pattern is the foundation of many data lakes on AWS.

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So what is AWS Glue? It's a fully managed, serverless data integration service. It helps you discover, prepare, and move data for analytics and machine learning. Serverless means there's no infrastructure to set up, and you pay only for the resources your jobs use while they run. Glue is made up of a few main pieces: the Data Catalog, crawlers, ETL jobs, and Glue Studio, a visual editor for building jobs. We'll go through each one.

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The Glue Data Catalog is a central metadata store. It organizes information into databases, and each database holds tables. A table records the schema, meaning the column names and data types, plus where the data lives, like an S3 path, and the format of the files. The key point is that the catalog stores metadata only. Your data stays in S3. And because the catalog is shared, services like Athena, EMR, Redshift Spectrum, and Glue ETL jobs can all use the same table definitions.

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Now, who fills the catalog? That's the crawler. A crawler works in four steps. First, you point it at a data store, such as an S3 path, a JDBC database, or DynamoDB. Second, it samples the data, using classifiers to recognize the format and infer the schema. Third, it creates or updates tables in the Data Catalog, and it can detect partitions, like folders organized by date. Fourth, you can run it on demand or on a schedule, so new data and schema changes are picked up automatically.

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The next piece is Glue ETL jobs, where ETL means extract, transform, and load. Glue runs your code on managed Apache Spark, so you write a PySpark or Scala script and Glue handles the workers. If you'd rather not code, Glue Studio lets you build jobs visually by dragging and dropping. Triggers and workflows let you run jobs on a schedule, on an event, or chain several jobs together. Typical tasks include converting CSV files to Parquet, cleaning data, joining datasets, and preparing features for machine learning.

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Now Amazon Athena. Athena is an interactive query service that lets you run standard SQL directly on data in S3. It's serverless, so there's nothing to set up or manage. You just write a query and get results. Under the hood it's built on the open source Trino engine. Athena uses the Glue Data Catalog to know what tables exist, and it saves query results to an S3 location. You pay based on the amount of data each query scans, which makes how you store your data really matter.

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Because Athena charges by data scanned, the goal is to scan less. Use columnar formats like Parquet or ORC, so Athena reads only the columns your query needs. Partition your data, for example by date, so queries can skip folders they don't need. Compress your files to reduce the bytes scanned. And avoid SELECT star, naming only the columns you want. These habits can cut both cost and query time dramatically. This is where a Glue ETL job often helps, by converting raw CSV into partitioned Parquet.

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So when do you use which? If you need to discover the structure of data and add it to the catalog, use a Glue crawler. If you need to clean, convert, or move data at scale, or on a schedule, use a Glue ETL job. If you want quick, ad hoc SQL queries on data in S3, use Athena. And if you need table definitions shared across services, that's the Data Catalog. For example, you might crawl raw CSV, use a Glue job to convert it to Parquet, crawl again, and then query it with Athena.

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So to recap, AWS Glue is a serverless data integration service, with a Data Catalog for metadata, crawlers to discover schemas, and ETL jobs to transform data. Amazon Athena lets you query data in S3 with SQL, using the catalog, and you pay for what it scans. Together they take you from raw files to answers, without managing servers. Thanks for watching, and I'll see you in the next one.